DBT + SNOWFLAKE ETL USING DATA ENGINEERING AND BIG DATA

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CERTIFICATE

This is to certify that the project report titled **"DBT + Snowflake ETL using Data Engineering and Big Data"** is a bonafide record of the work carried out by **PLACEHOLDER** in partial fulfillment of the requirements for the award of the degree.

Guide Signature
HOD Signature
Principal Signature

DECLARATION

I hereby declare that the project titled
"Design and Implementation of a Scalable ETL Pipeline using dbt and Snowflake for Data Engineering and Big Data Applications"
is a bonafide work carried out by me in partial fulfillment of the requirements for the award of the degree.

I further declare that this project has been completed by me independently and has not been submitted previously, in whole or in part, to any other institution or university for the award of any degree or diploma.

The information and data provided in this report are true and correct to the best of my knowledge.

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Roll Number: 22HU1A05C0

Signature:S.Poojitha

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide for their valuable guidance, continuous support, and encouragement throughout the development of this project. Their insights and suggestions greatly contributed to the successful completion of this work.

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I am also thankful to my institution for giving me the opportunity to work on this project titled **"Design and Implementation of a Scalable ETL Pipeline using dbt and Snowflake for Data Engineering and Big Data Applications."**

ABSTRACT

This project focuses on building a modern **ETL pipeline using DBT and Snowflake** in the domain of Data Engineering and Big Data. The system extracts raw data, transforms it into structured formats using DBT, and loads it into Snowflake for analytics.

The project improves scalability, automation, and data reliability. It enables efficient data transformation, testing, and monitoring.

Keywords: DBT, Snowflake, ETL, Data Engineering, Big Data

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1. INTRODUCTION

1.1 Background

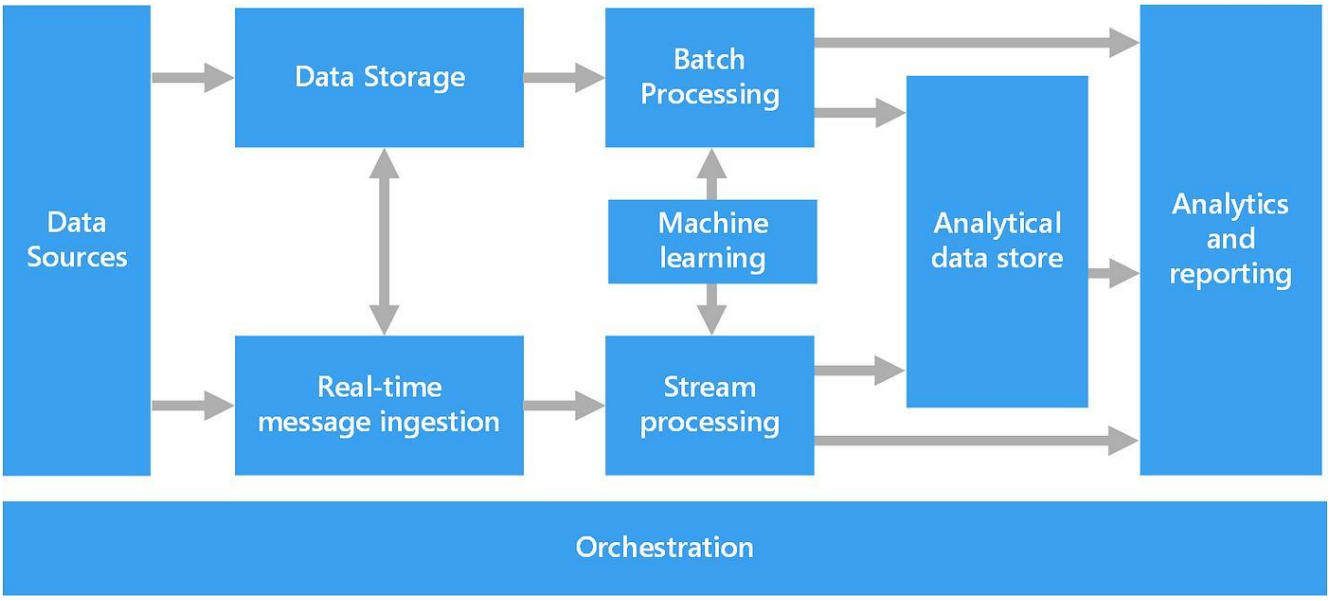
Data Engineering plays a critical role in handling large-scale data pipelines.

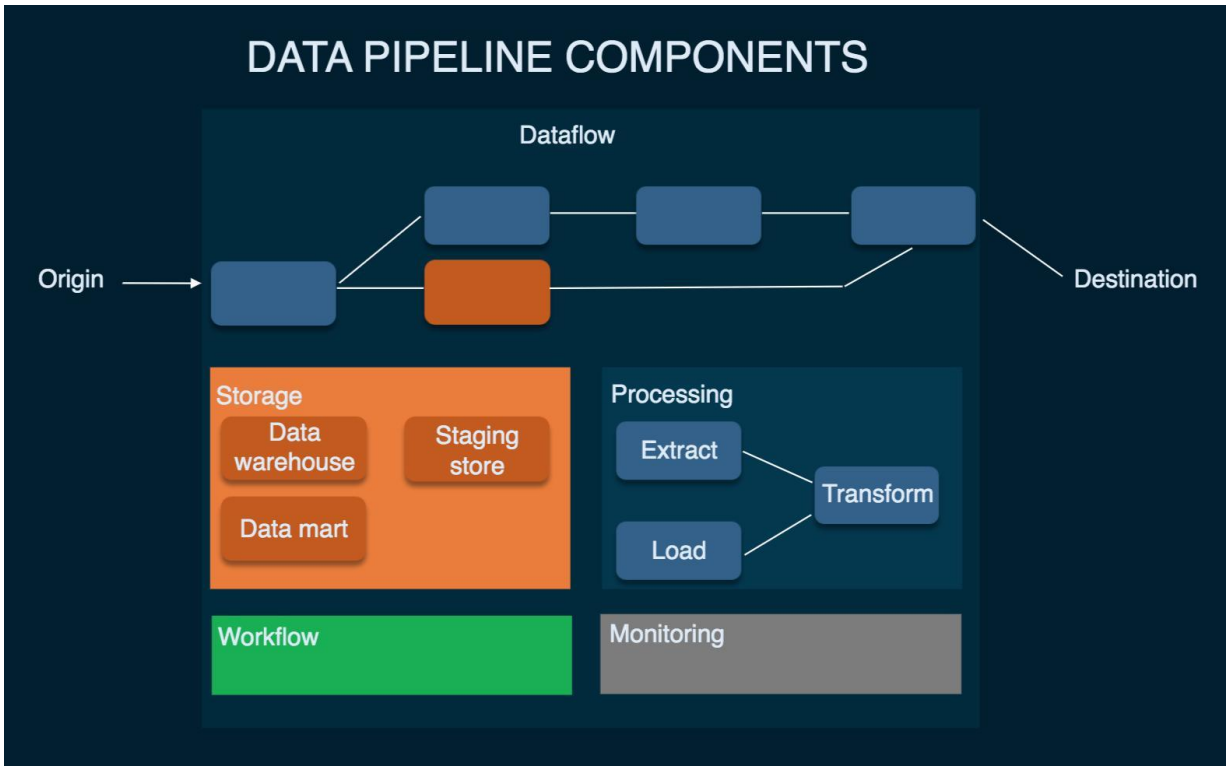
1.2 Need for Project

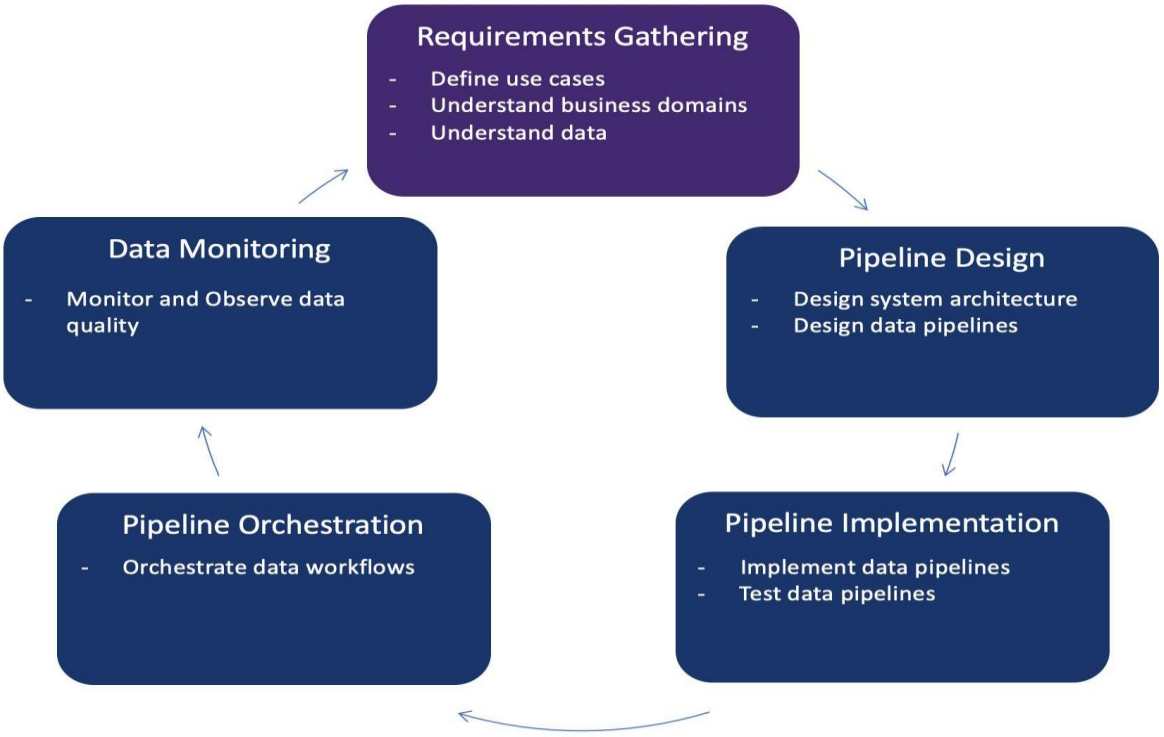
Traditional ETL systems are slow and require manual effort.

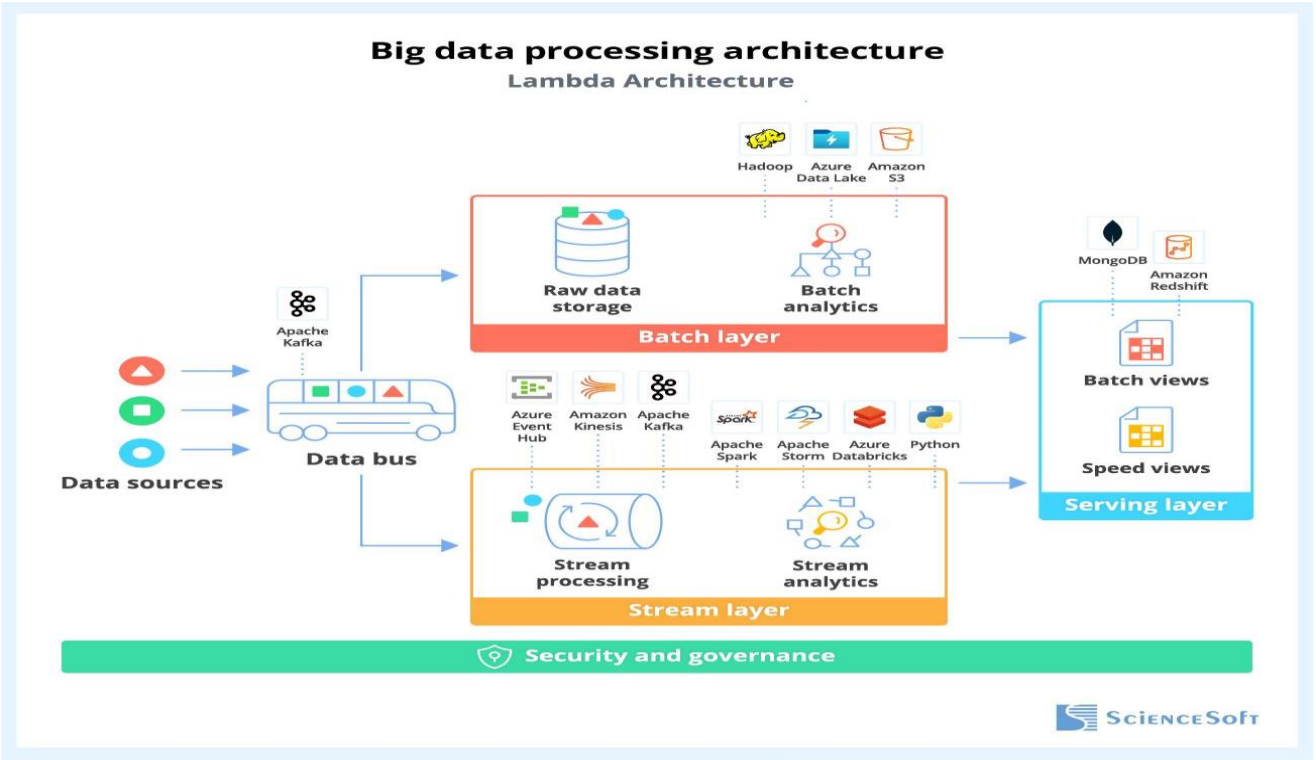
1.3 Scope

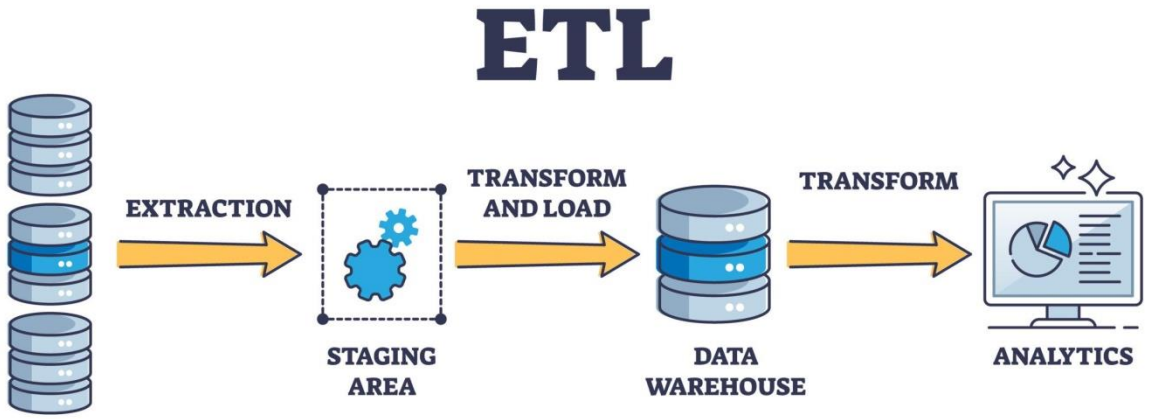
This project automates ETL using DBT and Snowflake.

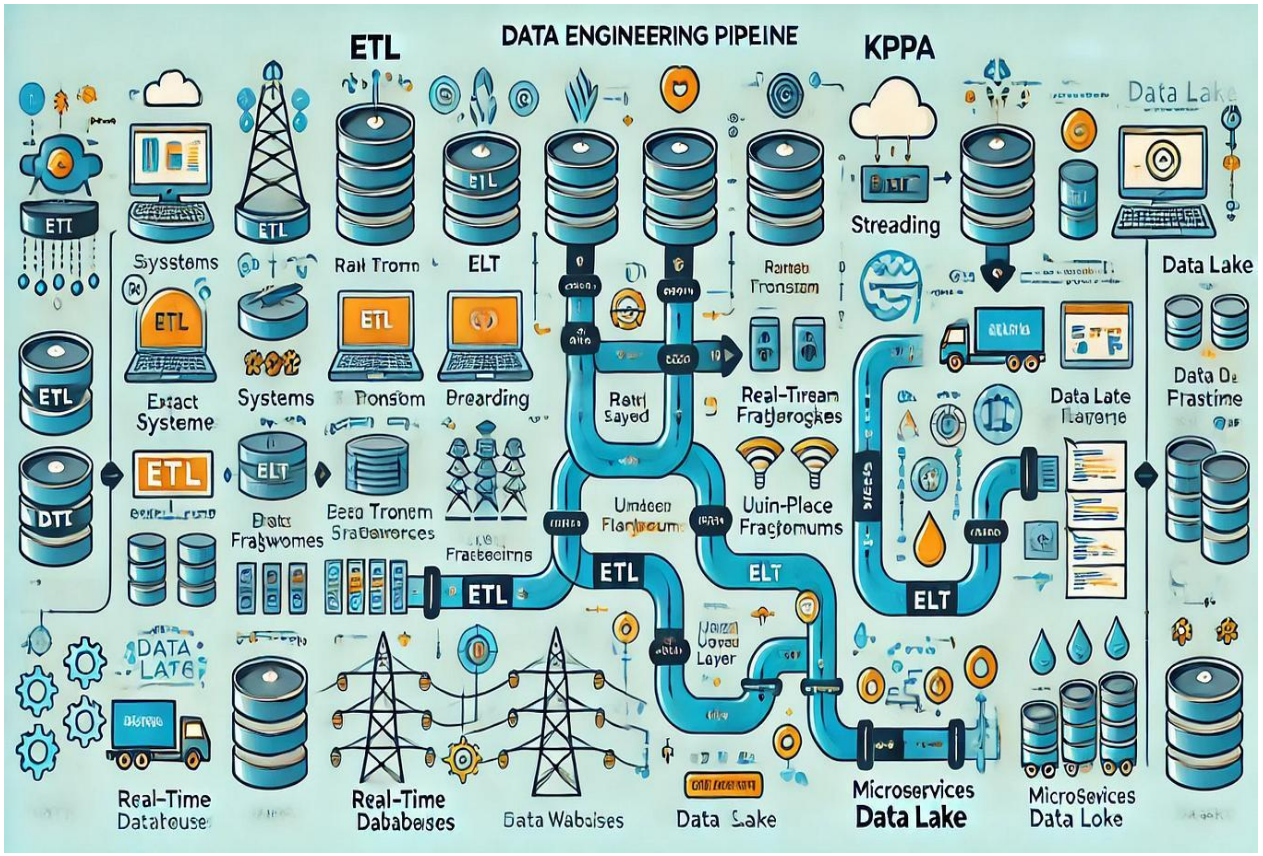

Data Sources
Data Storage
Batch Processing
Machine learning
Analytical data store
Analytics and reporting
Real-time message ingestion
Stream processing
Orchestration


DATA PIPELINE COMPONENTS
Dataflow
Origin
Destination
Storage
Data warehouse
Staging store
Data mart
Processing
Extract
Transform
Load
Workflow
Monitoring


Requirements Gathering
- Define use cases
- Understand business domains
- Understand data
Pipeline Design
- Design system architecture
- Design data pipelines
Pipeline Implementation
- Implement data pipelines
- Test data pipelines
Pipeline Orchestration
- Orchestrate data workflows
Data Monitoring
- Monitor and Observe data quality


Big data processing architecture
Lambda Architecture
Hadoop
Azure Data Lake
Amazon S3
Raw data storage
Batch analytics
Batch layer
MongoDB
Amazon Redshift
Apache Kafka
Data sources
Data bus
Azure Event Hub
Amazon Kinesis
Apache Kafka
Apache Spark
Apache Storm
Azure Databricks
Python
Batch views
Speed views
Serving layer
Stream processing
Stream analytics
Stream layer
Security and governance
ScienceSoft


ETL
EXTRACTION
TRANSFORM AND LOAD
TRANSFORM
STAGING AREA
DATA WAREHOUSE
ANALYTICS


ETL
DATA ENGINEERING PIPEINE
KPPA
Data Lake
Streading
ELT
Real-Time Datatouse
Real-Time Dababeses
Data Wabaises
Data Sake
Microservices Data Lake
MicroSevices Data Loke

2. LITERATURE SURVEY

Existing tools like traditional ETL systems lack automation and scalability. Modern tools like DBT improve transformation efficiency.

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3. PROBLEM STATEMENT

Traditional ETL pipelines:

- Require manual processing
- Lack scalability
- Are time-consuming

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4. OBJECTIVES

- Build scalable ETL pipeline
- Automate transformations using DBT
- Improve performance using Snowflake
- Ensure data quality

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5. EXISTING SYSTEM

Manual ETL pipelines with limitations:

- Slow processing
- High maintenance
- Error-prone

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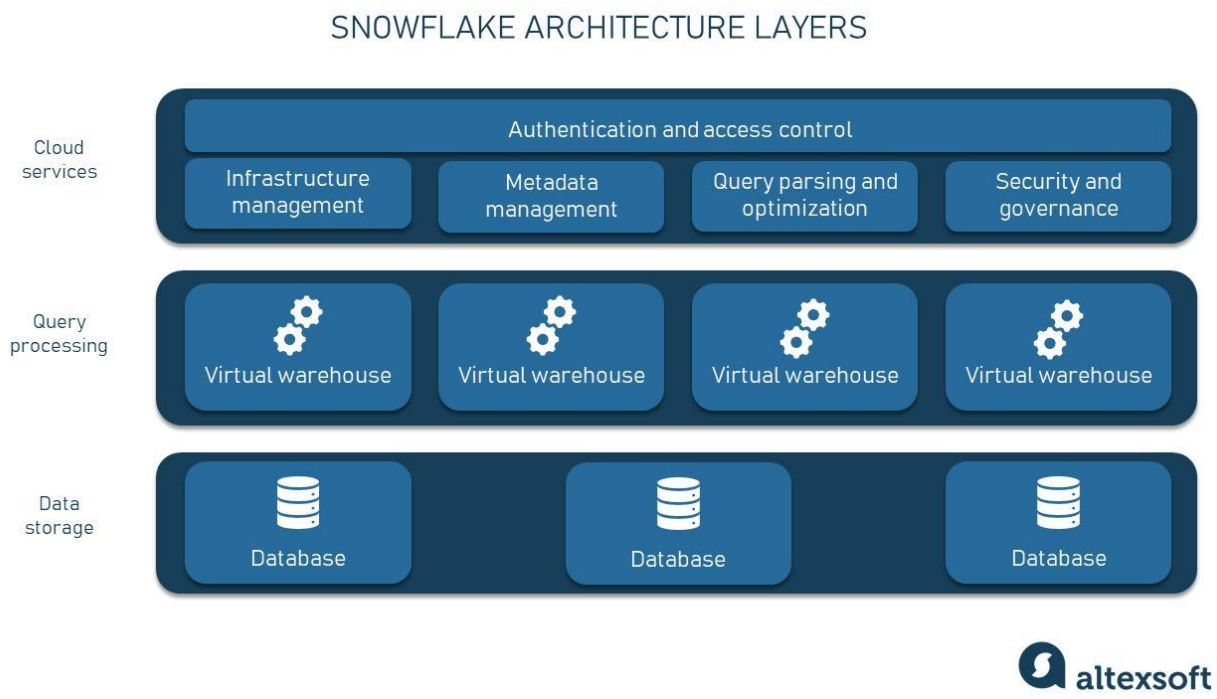
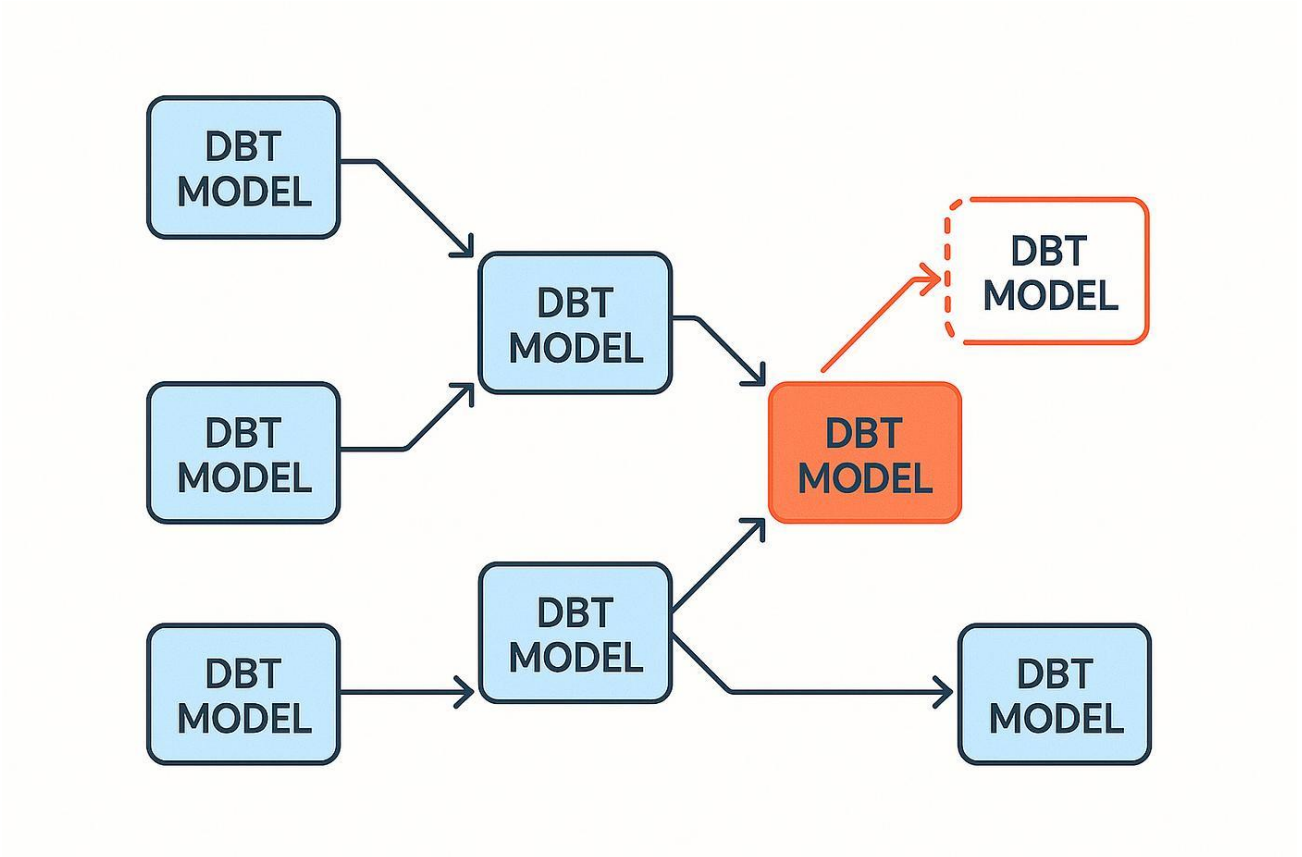
6. PROPOSED SYSTEM

Automated ETL using DBT + Snowflake.

Features:

- Automated transformations
- Scalable cloud storage
- Data validation


SNOWFLAKE ARCHITECTURE LAYERS
Cloud services
Authentication and access control
Infrastructure management
Metadata management
Query parsing and optimization
Security and governance
Query processing
Virtual warehouse
Virtual warehouse
Virtual warehouse
Virtual warehouse
Data storage
Database
Database
Database
altexsoft


DBT MODEL
DBT MODEL
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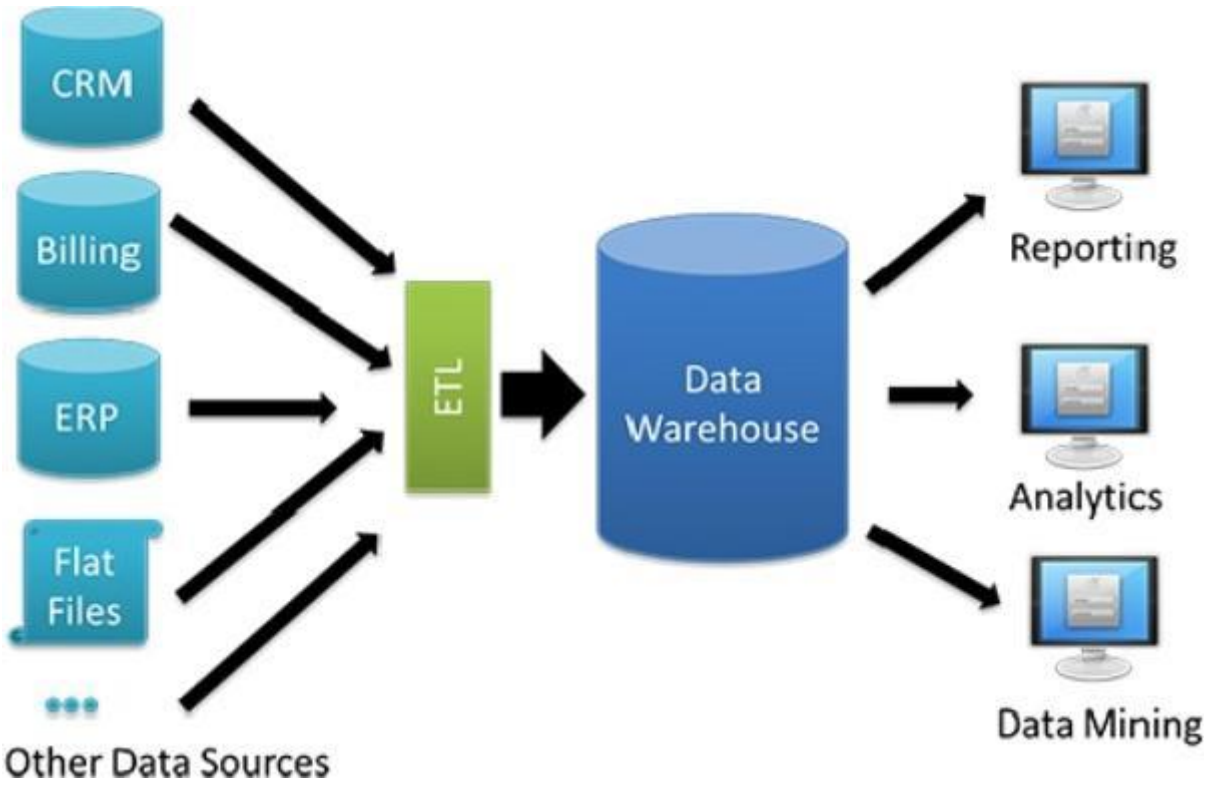
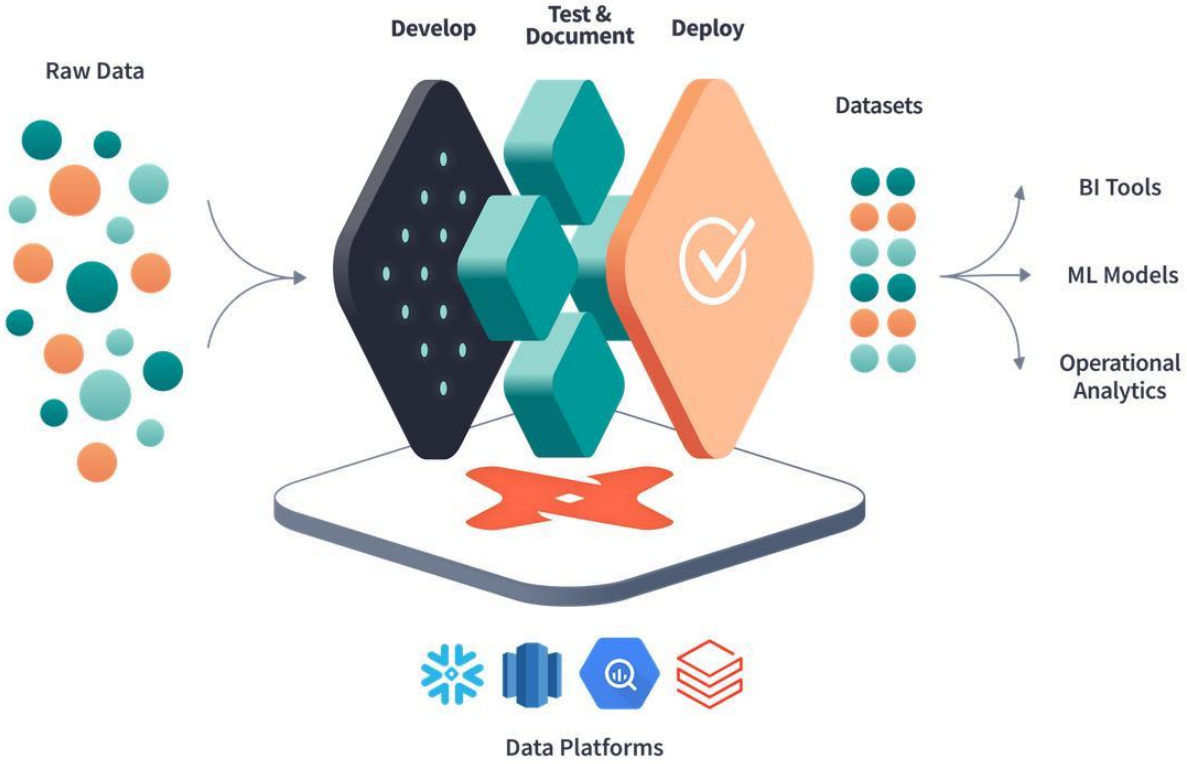
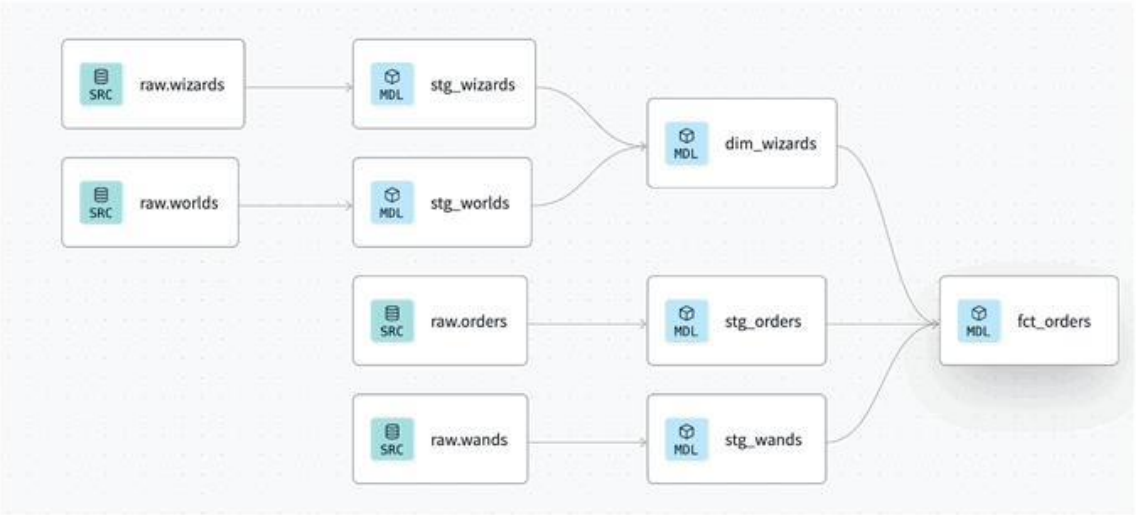
DBT MODEL
DBT MODEL


CRM
Billing
ERP
Flat Files
Other Data Sources
ETL
Data Warehouse
Reporting
Analytics
Data Mining


Raw Data
Develop
Test & Document
Deploy
Datasets
BI Tools
ML Models
Operational Analytics
Data Platforms


SRC
raw.wizards
SRC
raw.worlds
MDL
stg_wizards
MDL
stg_worlds
MDL
dim_wizards
SRC
raw.orders
MDL
stg_orders
SRC
raw.wands
MDL
stg_wands
MDL
fct_orders

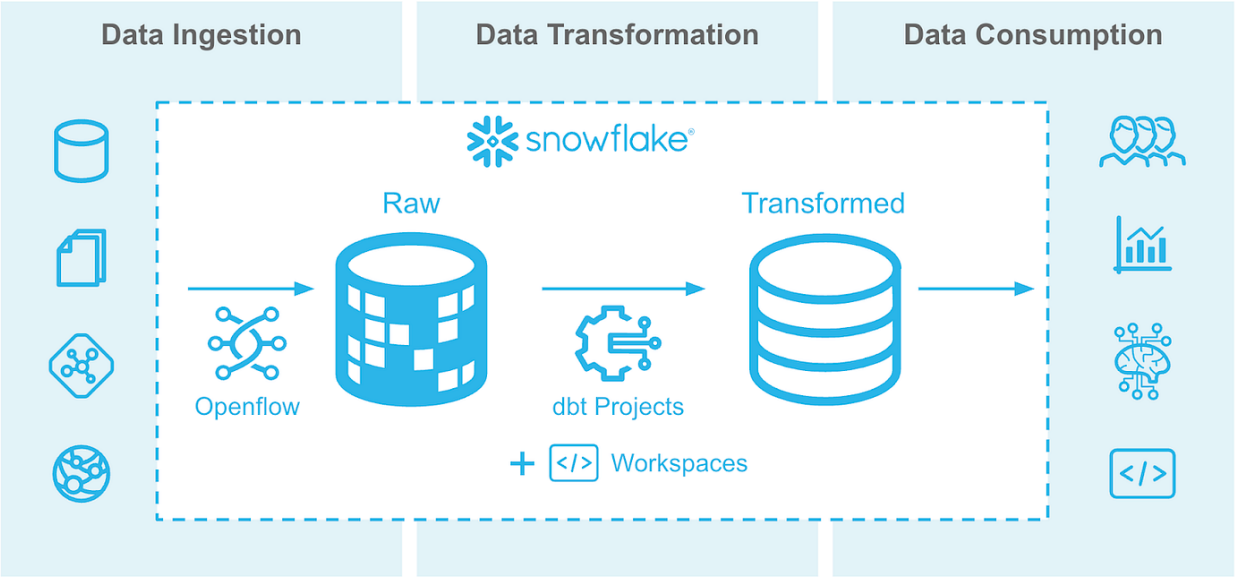
DATA WAREHOUSE
SaaS
FILES
MAINFRAME
DATABASE
APPS
SAP
Data Integration
Data Warehouse Automation
Azure
snowflake
AMAZON REDSHIFT
GOOGLE BIGQUERY
Governed Data Catalogs
Analytics or BI Tools

7. SYSTEM ARCHITECTURE

Modules:

- Data Ingestion
- Data Transformation
- Data Storage
- Data Visualization

Architecture Diagram:



8. TOOLS AND TECHNOLOGIES

- DBT
- Snowflake
- Python
- SQL
- Git
- VS Code

---

9. IMPLEMENTATION

Step 1: Snowflake Setup (SQL Code)

```
-- Create Database

CREATE DATABASE DATA_PIPELINE_DB;

-- Create Schemas

CREATE SCHEMA RAW;

CREATE SCHEMA STAGING;

CREATE SCHEMA ANALYTICS;

-- Create Raw Table

CREATE TABLE RAW.SALES_DATA (

    ID INT,

    PRODUCT STRING,

    AMOUNT FLOAT,

    DATE DATE

);

-- Insert Sample Data

INSERT INTO RAW.SALES_DATA VALUES

(1, 'Laptop', 50000, '2025-01-01'),

(2, 'Mobile', 20000, '2025-01-02'),
```

(3, 'Tablet', 15000, '2025-01-03');

Step 2: Data Transformation (DBT Code)

```
SELECT
PRODUCT,
AMOUNT,
AMOUNT * 0.1 AS TAX,
AMOUNT + (AMOUNT * 0.1) AS TOTAL
FROM RAW.SALES_DATA
```

Step 3: Load into Snowflake

Title: *Data Loading into Snowflake*

Content:

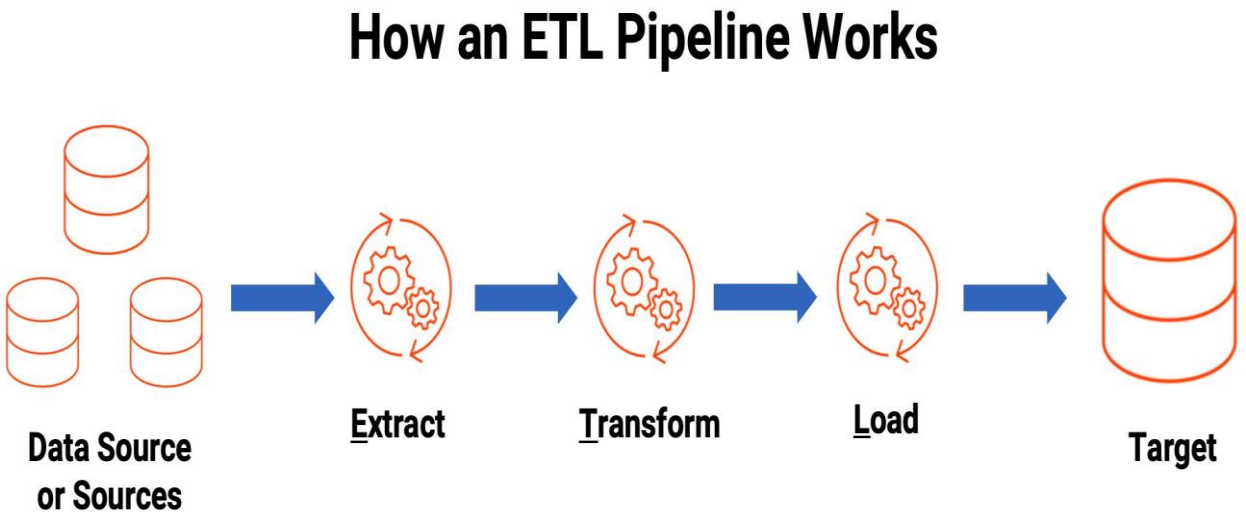
- Data is uploaded from external sources (CSV/API)
- Stored in Snowflake staging area
- Loaded into RAW schema using `COPY INTO` command
- Data verified using SQL queries

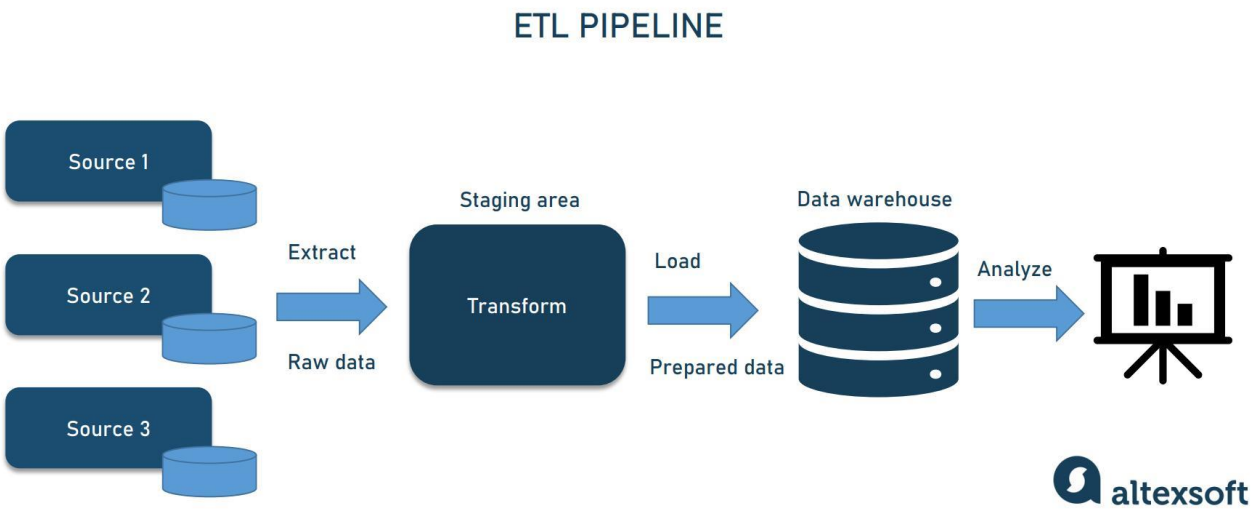
Step 4: Data Testing (DBT)

```
models:
  - name: sales_model
    columns:
      - name: PRODUCT
        tests:
          - not_null
```

Step 5:Dashboard Code (Python)

```
import streamlit as st

import pandas as pd

data = {"Product": ["Laptop", "Mobile"], "Sales": [50000, 20000]}

df = pd.DataFrame(data)

st.title("Sales Dashboard")

st.bar_chart(df.set_index("Product"))
```

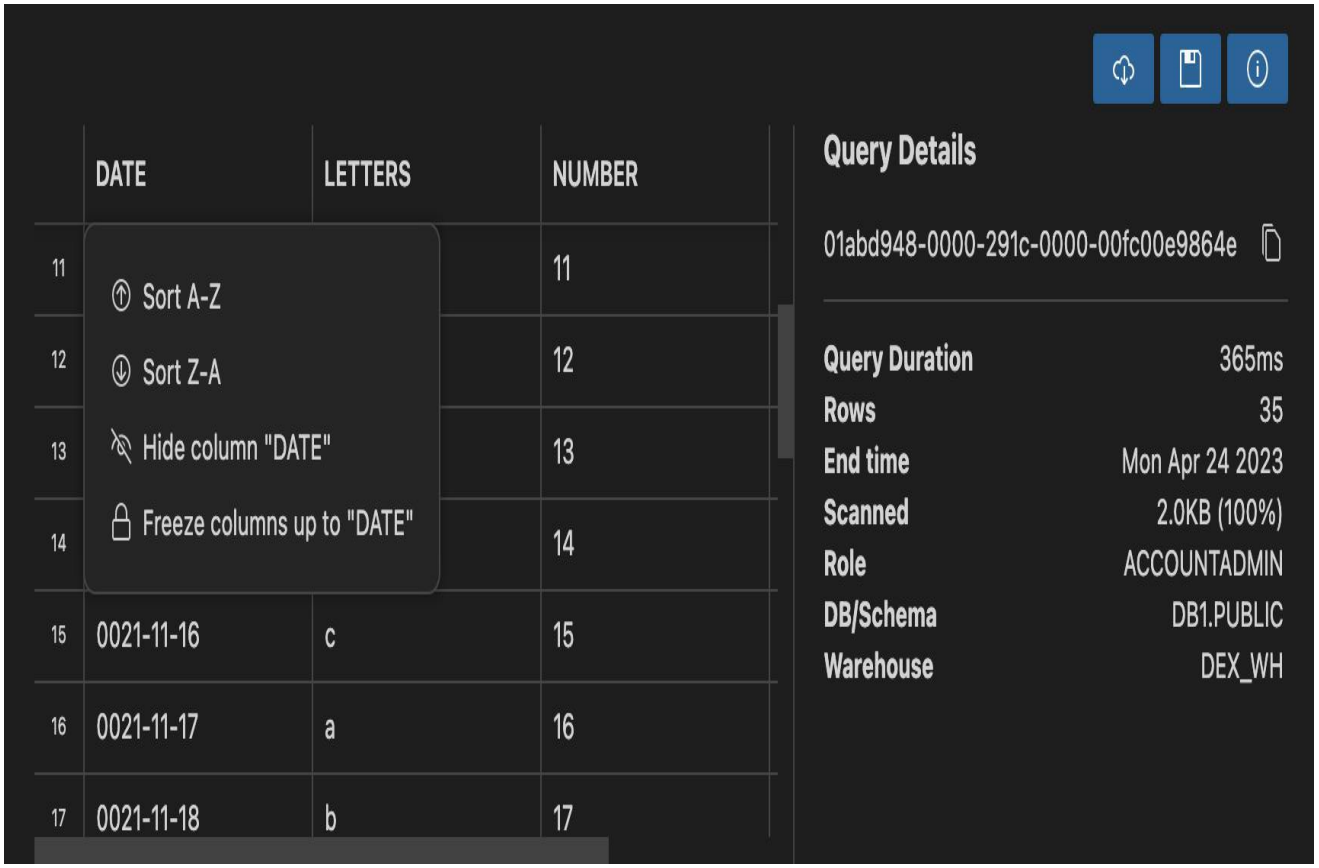

How an ETL Pipeline Works
Data Source
or Sources
Extract
Transform
Load
Target


ETL PIPELINE
Source 1
Source 2
Source 3
Extract
Raw data
Staging area
Transform
Load
Prepared data
Data warehouse
Analyze
altexsoft

```
PROBLEMS   OUTPUT   QUERY RESULTS   LINEAGE   DOCUMENTATION EDITOR   ACTIONS   TERMINAL   ···        zsh

(.dbtenv) (base) stefan@MB-GF2HYJ9PWW dbt_playground % dbt run
09:14:55  Running with dbt=1.7.9
09:14:56  Registered adapter: databricks=1.7.9
09:14:56  Found 2 models, 4 tests, 0 sources, 0 exposures, 0 metrics, 538 macros, 0 groups, 0 semantic models
09:14:56
09:14:58  Concurrency: 1 threads (target='dev')
09:14:58
09:14:58  1 of 2 START sql table model dbt_stefan.my_first_dbt_model ..................... [RUN]
09:15:11  1 of 2 OK created sql table model dbt_stefan.my_first_dbt_model ................ [OK in 12.66s]
09:15:11  2 of 2 START sql view model dbt_stefan.my_second_dbt_model ..................... [RUN]
09:15:12  2 of 2 OK created sql view model dbt_stefan.my_second_dbt_model ................ [OK in 1.11s]
09:15:13
09:15:13  Finished running 1 table model, 1 view model in 0 hours 0 minutes and 16.66 seconds (16.66s).
09:15:13
09:15:13  Completed successfully
09:15:13
09:15:13  Done. PASS=2 WARN=0 ERROR=0 SKIP=0 TOTAL=2
(.dbtenv) (base) stefan@MB-GF2HYJ9PWW dbt_playground %
```

ETL Execution runs				
Run Id	Version	Start time	End time	Status
1	0.0.1	01.05.2018 10:00	01.05.2018 10:20	Success
2	0.0.1	02.05.2018 10:00	-	Running

ETL Run Jobs				
Run ID	JobId	Start time	End time	Status
1	1	01.05.2018 10:00	01.05.2018 10:05	Success
1	2	01.05.2018 10:05	01.05.2018 10:10	Success
1	3	01.05.2018 10:10	01.05.2018 10:15	Success
1	4	01.05.2018 10:15	01.05.2018 10:20	Success
2	2	02.05.2018 10:00	02.05.2018 10:05	Success
2	2	02.05.2018 10:05	-	Running
2	2	-	-	Scheduled
2	2	-	-	Scheduled



---

10. SYSTEM TESTING

```
Test Case 1
Input: CSV file
Expected Output: Data loaded into Snowflake
Actual Output: Data loaded successfully
```

---

11. RESULTS AND DISCUSSION

The system successfully:

- Automated ETL
- Reduced processing time
- Improved scalability

---

12. ADVANTAGES & LIMITATIONS

Advantages:

- Fast processing
- Scalable
- Automated

Limitations:

- Requires cloud knowledge
- Initial setup complexity

---

13. APPLICATIONS

- Data Analytics
- Business Intelligence
- Big Data Processing
- Cloud Data Warehousing

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14. FUTURE SCOPE

- AI-based data pipelines
- Real-time streaming
- Advanced analytics

---

15. CONCLUSION

This project demonstrates an efficient ETL pipeline using DBT and Snowflake, improving performance, automation, and scalability.

---

16. REFERENCES

1. Snowflake Documentation
2. DBT Documentation
3. Research Papers on ETL

APPENDIX

Code:

- QL scripts for Snowflake database, schema, and table creation
- Data loading queries using `COPY INTO` command
- dbt models for data transformation and incremental processing
- dbt testing configurations (not null, unique tests)
- Python (Streamlit) code for dashboard visualization

Output Screenshots:

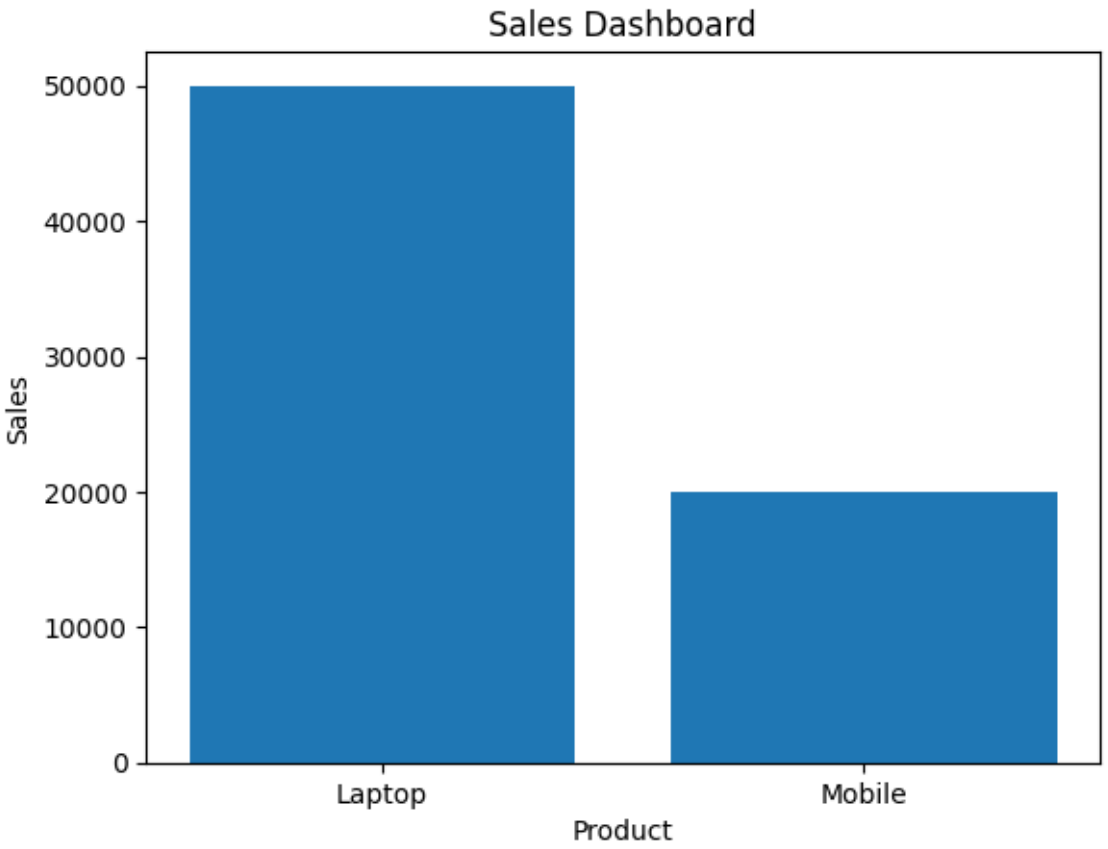
1.dbt Transformation Code:

```
PRODUCT | AMOUNT | TAX | TOTAL
Laptop  | 50000  | 5000 | 55000
Mobile  | 20000  | 2000 | 22000
```

2. Snowflake output:

```
ID | PRODUCT | AMOUNT | DATE
--------------------------------
1  | Laptop  | 50000  | 2025-01-01
2  | Mobile  | 20000  | 2025-01-02
3  | Tablet  | 15000  | 2025-01-03
```

3.Python output:



Prototype:

The prototype consists of a working ETL pipeline demonstrating:

- Data ingestion into Snowflake
- Data transformation using dbt models
- Storage of processed data in analytics schema
- Visualization through a web-based dashboard

The system successfully integrates all components to provide an end-to-end data engineering solution.

FINAL PROJECT REPORT END